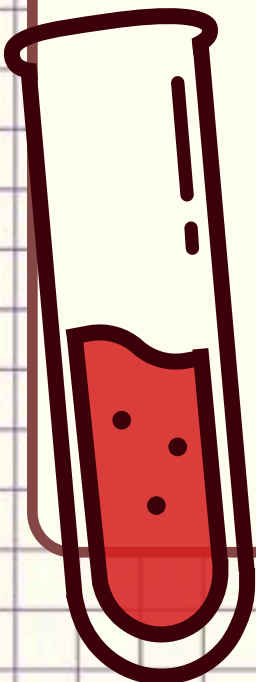




Presenting Science

miniGEM - 2025





You've put in a lot of hard work, researching and developing your miniGEM ideas and now it's time to tell the judges about it! Presenting science is an important part of every scientist's journey. It's the moment where your cool ideas actually become reality, because the world gets to know what you have been up to. However, it can also be a tricky process.

This is a step-by-step guide to help you communicate your cool ideas clearly for others to understand. However, it is only a guide. Feel free to try other strategies that are easier and that may be more intuitive. The only rule that is an absolute must when presenting science is to make sure that your audience understands your message clearly. As long as you tick that box, everything else is secondary. But in case you'd like some, here are some of our favorite tips and tricks!



1. Summarize your entire idea into just one sentence.

You might have come across the idea of a topic/thesis statement in your class. That is what this section is about. Think about newspaper headings or YouTube video titles or the first sentence in a good book. To grab the attention of your audience, you need to be able to communicate your entire idea in just one sentence

2. Give an Elevator Pitch.

I) What are all the important parts of your idea? In other words, why should anyone care about this thing that you have just built? This section is still not science-heavy, however it can include scientific principles as long as it is clear.



II) To do this properly you should first make an outline of all the key things you want people to know. Depending on the length of your presentation, this outline can vary in length. For miniGEM, we recommend having around 3 main points.

III) Take that outline and make it into your pitch. At the very minimum, whenever you're presenting science, you are trying to convince your reader/listener that what you've come up with is important. So don't be afraid to be persuasive in this section

3. Time For Details!

Now is the time to show off all the cool synthetic biology you've learned!

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I) Your idea thus far sounds cool and interesting. However, it is currently a ball of fluff. To give it more substance, we need facts. What makes you think this idea will work? Why is it needed? Has anyone done anything similar? Stats, proteins, organisms, techniques. As much as you know, now is this the time to share.

*Be careful in this section! We're asking you to tell us everything you know, but it should still be relevant. If you start listing details without connecting it back to your main idea, you will lose your audience.

II) Our recommendation is, first explain the general science behind your idea. The proteins, organisms, and techniques you're using. Then go over each point mentioned in your elevator pitch and support them with facts and science. This will keep this section focused and engaging.

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III) Visuals! Visuals! Visuals! You can probably see your idea and your proteins super clearly in your mind, however the judges are not in your mind, so show them. We recommend making a sketch of your system and how it might be implemented.

IV) This section should be the longest part of your presentation. However, do not stress about needing to know or say everything possible. miniGEM focuses on the clarity of your idea and how well developed and thought out your creations are. You do not need to be an expert by any means!

4. Special Prizes

If you've decided to compete for either the entrepreneurship or sustainability special prizes, now is the time to tell us!

I) This is comparable to the discussion section of most papers, where the idea is to make your creation relevant to the real world. Did you create something that could be a really cool new product? Tell that to your audience! Does your idea strongly support one of the United Nations 17 Sustainable Development Goal? People want to know! Doing science just for the sake of science is cool, but doing so in a way that is applicable to the world is way cooler. We strongly encourage you to think about this aspect of your creation and to apply for a special prize!



5. End on a good note!

People often only remember the very beginning and the very end of things, so let's finish off strong! You've told us a lot of cool things, but we might have forgotten, so give us a reminder of the most important points. Our recommendation is go back to your elevator pitch. But don't repeat it word for word. Remix it using some of the science background that we now know. Keep it short. We also recommend ending on the big picture of what your idea is, what it does and most especially, why it's important.

Overall, we recommend to start broad, zoom into the details and then finish with a slow cinematic zooming out. This gets your audience engaged from the start, tells all the important science, and sends them off with the same awe and satisfaction that you feel when the credits start rolling after a really good movie. Good luck!